uPrime

Because u' matters

User Manual

Version 0.3 · Alpha Release

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github.com/CmdrRyder/uPrime

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Chapter 1

Introduction

uPrime is an open-source desktop application for post-processing planar and stereo PIV (Particle Image Velocimetry) data. It is designed for researchers who want to perform standard turbulence analysis quickly, without writing code for every dataset. All analyses run in separate windows so that you can keep several results open side by side.

1.1 What uPrime can do

- Load multi-snapshot DaVis Tecplot `.dat` files produced by LaVision DaVis.
- Display mean velocity fields, standard deviation, vorticity, and velocity vectors.
- Compute Reynolds stresses and extract line profiles across any direction.
- Compute and visualise the turbulent kinetic energy (TKE) budget.
- Perform spatial, temporal, and space–time spectral analysis.
- Analyse turbulence anisotropy via Lumley triangle and barycentric maps (stereo only).
- Compute two-point spatial and temporal correlations and integral length/time scales.
- Perform Proper Orthogonal Decomposition (POD) and reconstruct the flow field.
- Correct camera calibration misalignment with the coordinate transform tool (rotation and origin shift).
- Export all results to Tecplot `.dat` or `.csv` for further analysis.

1.2 System requirements

Table 1.1: Minimum requirements to run uPrime.

Item	Requirement
Operating system	Windows 10/11, macOS 12+, or Linux (Ubuntu 20+)
Python	3.10 or later
RAM	8 GB minimum; 16 GB recommended for large datasets
Display	1920×1080 or larger strongly recommended

1.3 Installation

1. Clone or download uPrime from github.com/CmdrRyder/uPrime.
2. Install the required Python packages:

   ```
   pip install -r requirements.txt
   ```

3. Launch the application:

```
python main.py
```

A standalone `.exe` for Windows is available on the GitHub Releases page and requires no Python installation.

> **Note**
>
> If you are using the `.exe`, simply double-click it to launch. No installation is needed.

Chapter 2

Main Window

When uPrime starts you will see the welcome screen (Figure 2.1), followed by the main window once data is loaded (Figure 2.2). The interface is divided into two areas: a **left sidebar** with all controls and a **right plot area** showing the velocity field.

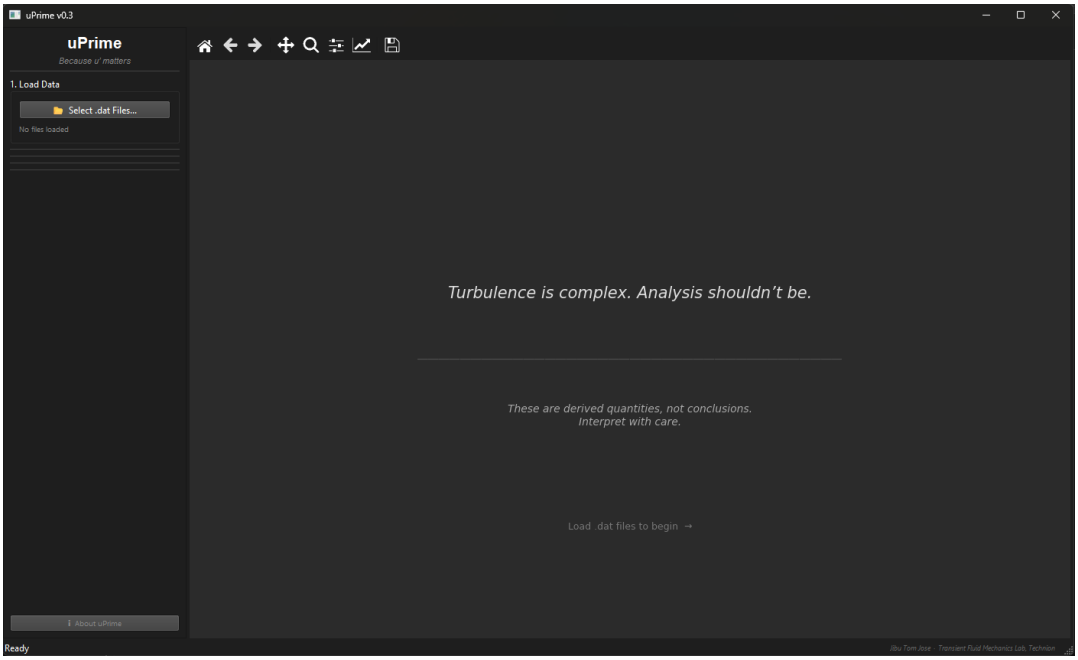

Figure 2.1: uPrime welcome screen. Load `.dat` files using the sidebar to begin.

2.1 Left sidebar

The sidebar is organised into numbered groups that are meant to be used in order from top to bottom.

2.1.1 1. Load Data

Click **Select .dat Files...** to open a file browser. Select one or more Tecplot `.dat` files exported from DaVis. You can select multiple files at once (hold `Ctrl` or `Shift`). uPrime loads all snapshots from the selected files and stacks them into a single dataset.

> **Note**
>
> uPrime expects the standard DaVis Tecplot format with variables `x`, `y`, `U`, `V` (and optionally `W` for stereo). Files from other sources may not load correctly.

The status bar at the bottom shows loading progress. Once loading is complete, the dataset information panel (Section 2.1.3) is populated automatically.

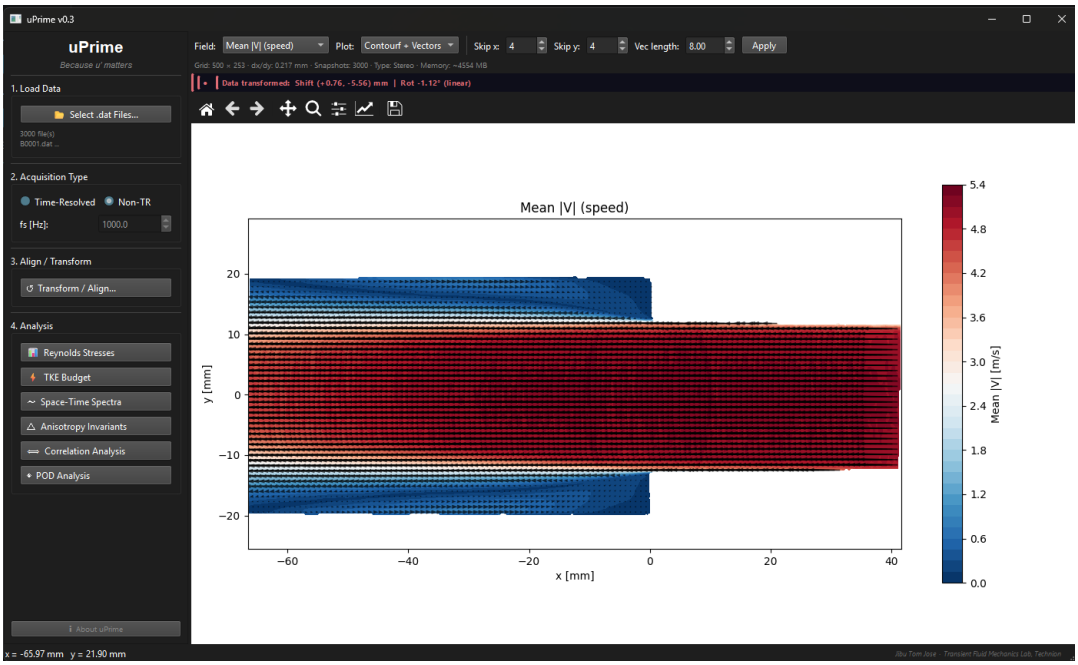

Figure 2.2: Main window showing the left sidebar and the mean speed field with velocity vectors overlaid.

2.1.2 2. Acquisition Type

Choose whether your data is **Time-Resolved (TR)** or **Non-TR**:

- **Time-Resolved**: the snapshots were acquired at a fixed sampling rate. Enter the sampling frequency f_s in Hz. This enables the temporal correlation and temporal spectral analysis tabs.
- **Non-TR**: the snapshots are statistically independent (e.g., double-frame PIV). Temporal analyses are not available.

> **Warning**
>
> Setting the wrong acquisition type will not crash uPrime, but temporal results will be physically meaningless. Always set f_s correctly before opening any analysis window.

2.1.3 3. Dataset Info

Displays a summary of the loaded dataset:

- **Grid**: number of vectors in x and y.
- **dx / dy**: spatial resolution in mm.
- **Snapshots**: total number of time steps loaded.
- **Type**: 2D or Stereo PIV.
- **Memory**: estimated RAM usage.

2.1.4 4. Analysis

Each button opens the corresponding analysis in a new window. Multiple windows can be open at the same time. See Chapters 3–8 for details on each module.

2.1.5 5. Align / Transform

Opens the coordinate transformation tool. See Chapter 9.

2.2 Field viewer

The right area displays the currently selected velocity field. Use the toolbar above the plot to zoom, pan, and save the figure.

2.2.1 Display options

The strip above the field viewer provides the following controls:

Table 2.1: Field viewer display controls.

Control	Description
Field	Choose which quantity to display (Mean U, Mean V, Std U, etc.)
Plot	Contourf only, vectors only, or both overlaid
Skip x / y	Spacing between displayed velocity vectors
Vec length	Scale multiplier for arrow length
Apply	Refresh the plot with current settings

Tip

For large grids, increase the Skip values to avoid a cluttered vector plot. A skip of 10–20 usually gives a clean result.

Chapter 3

Reynolds Stress Analysis

Open this window from **4. Analysis → Reynolds Stresses**.

Reynolds stresses are the time-averaged products of velocity fluctuations: $\langle u'u' \rangle$, $\langle v'v' \rangle$, $\langle w'w' \rangle$, $\langle u'v' \rangle$, $\langle u'w' \rangle$, $\langle v'w' \rangle$. The window shows 2D maps of each component and allows you to extract line profiles.

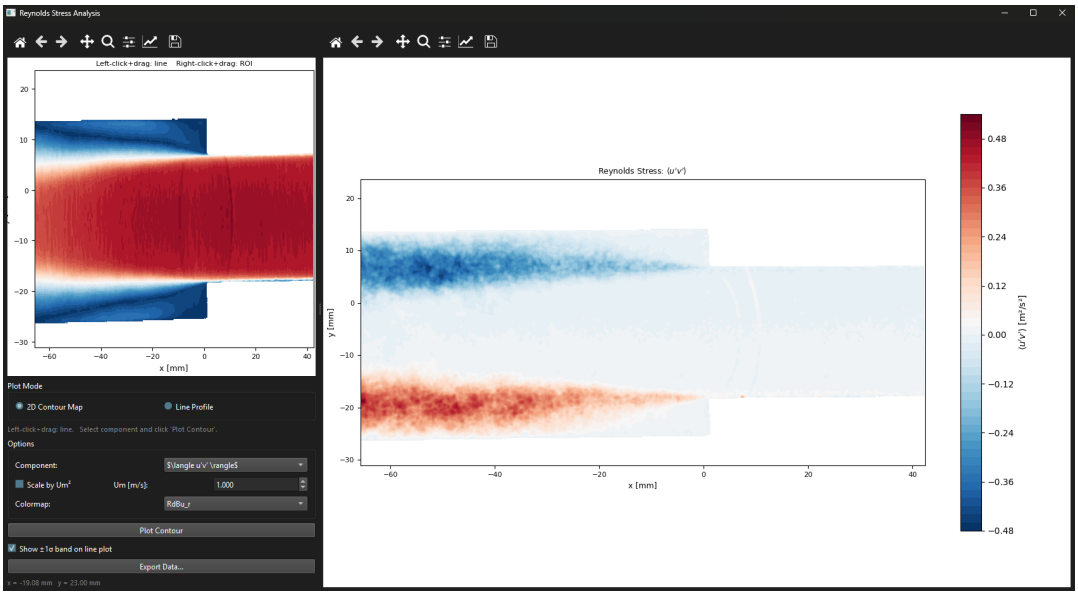

Figure 3.1: Reynolds stress window in 2D contour mode showing the $\langle u'v' \rangle$ component.

3.1 Selecting a component

Use the **Component** dropdown in the left panel to switch between stress components. For 2D PIV data, only $\langle u'u' \rangle$, $\langle v'v' \rangle$, and $\langle u'v' \rangle$ are available. All six components are available for stereo data.

3.2 Display mode

- **Contour**: shows the 2D filled contour map of the selected stress component. The colormap is symmetric about zero (`RdBu_r`).
- **Line Profile**: draw a line on the field to extract a 1D profile along that line. Left-click and drag to draw. The extracted profile is shown in the right panel.

3.2.1 Line profile options

When in line profile mode, three line orientations are available:

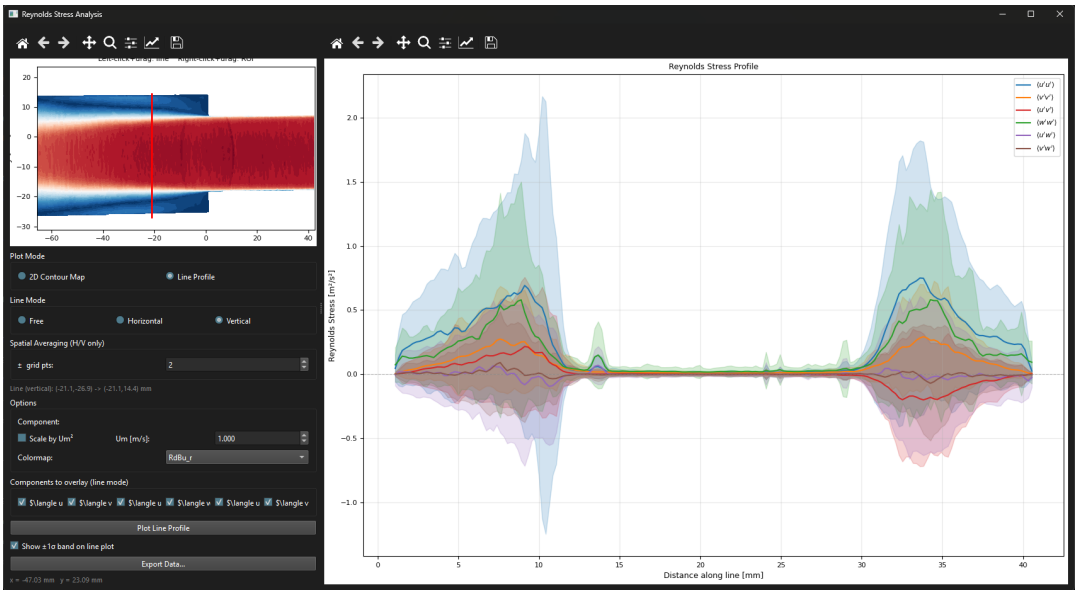

Figure 3.2: Reynolds stress window in line profile mode. All stress components are plotted simultaneously with $\pm 1\sigma$ uncertainty bands shaded.

- **Free**: any direction, interpolated along the drawn path.
- **Horizontal**: snaps the line to the nearest horizontal grid row.
- **Vertical**: snaps the line to the nearest vertical grid column.

The right panel shows all available stress components along the extracted line simultaneously, each in a different colour.

3.3 Standard deviation overlay

Check **Show $\pm 1\sigma$ band** to overlay a shaded uncertainty band on the line profile. This represents the standard deviation of the stress computed from the snapshot ensemble and gives a sense of statistical convergence.

3.4 Exporting

- **Export 2D Map...**: saves the current 2D stress field as a Tecplot `.dat` file.
- **Export Line Profile...**: saves the extracted profile as a `.csv` file.

Chapter 4

TKE Budget

Open this window from **4. Analysis → TKE Budget**.

The TKE budget window shows the spatial distribution of each term in the turbulent kinetic energy transport equation. The terms computed are:

Table 4.1: TKE budget terms available in uPrime.

Symbol	Name	Notes
k	TKE	Always available
P	Production	Always available
C	Convection	Always available
D	Turbulent diffusion	Always available
R	Residual	$R = -(P + C + D + \partial k/\partial t)$
$\partial k/\partial t$	Unsteady term	TR data only

Note

The residual R implicitly contains pressure diffusion and viscous dissipation, which cannot be computed from planar PIV data alone. A large residual indicates either poor convergence, missing terms, or measurement noise.

4.1 Using the window

The left panel shows the mean field. The right panel shows the selected budget term as a 2D contour map or a line profile (Figure 4.1).

1. Select the budget term from the **Term** dropdown.
2. Choose **Contour** or **Line Profile** mode.
3. In line profile mode, left-click and drag on the left field to draw a profile line.
4. Export using the buttons in the top-right corner.

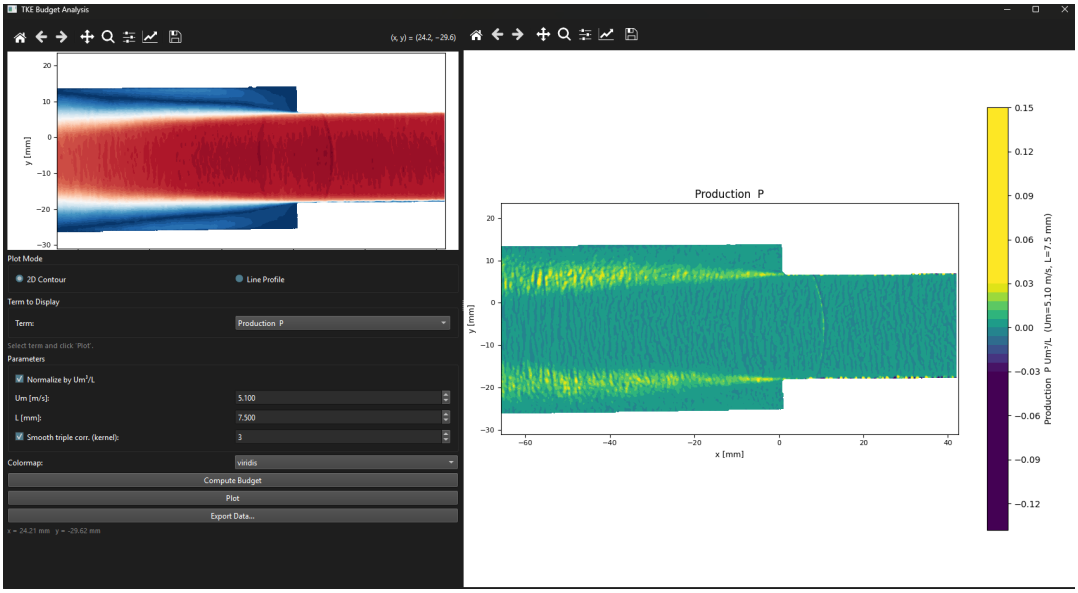

Figure 4.1: TKE Budget window showing the TKE production term P as a 2D contour map.

Chapter 5

Space–Time Spectral Analysis

Open this window from **4. Analysis → Space-Time Spectra**.

This window has three tabs. The **Spatial** tab is always available. The **Temporal** and **Spatiotemporal** tabs require time-resolved data.

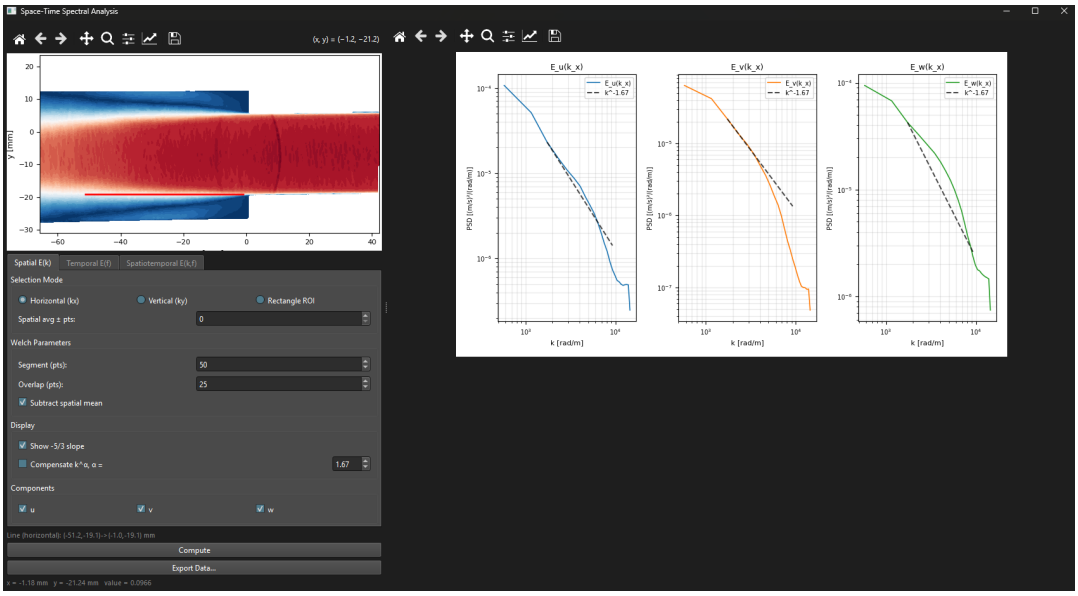

Figure 5.1: Space–Time Spectral Analysis window showing the spatial energy spectra $E(k_x)$ for all three velocity components along a vertical line, with the Kolmogorov $-5/3$ slope overlaid.

5.1 Spatial spectra (Tab 1)

Computes the one-dimensional spatial energy spectrum $E(k)$ along a user-drawn line or within a rectangular ROI.

1. Choose **Line** or **ROI** mode.
2. **Line mode**: left-click and drag on the field to draw a line. The spectrum is computed along that line.
3. **ROI mode**: right-click and drag to draw a rectangle. The spectrum is averaged over all lines within the ROI.
4. Select the velocity component (U or V).
5. Click **Compute Spatial Spectrum**.

The result is displayed as $E(k)$ vs wavenumber k on a log–log scale.

5.2 Temporal spectra (Tab 2, TR only)

Computes the power spectral density $E(f)$ at a selected point or averaged over an ROI using Welch's method.

1. Left-click on the field to pick a reference point, or draw an ROI.
2. Set the number of Welch segments using the **Segments** spinbox. More segments give a smoother spectrum but lower frequency resolution.
3. Click **Compute Temporal Spectrum**.

Tip

Use more segments for noisy data. Use fewer segments if you need to resolve low-frequency content.

5.3 Space–time spectra (Tab 3, TR only)

Computes the 2D space–time spectrum $E(k, f)$, which shows how energy is distributed across both spatial scales and frequencies simultaneously. This is useful for identifying convective structures (ridges of constant f/k give the convection velocity).

Set the analysis line direction and click **Compute**. The result is shown as a 2D colour map.

Chapter 6

Anisotropy Invariant Analysis

Open this window from **4. Analysis → Anisotropy Invariants**.

> **Note**
>
> This module requires **stereo PIV** data with all three velocity components (U, V, W). It is not available for 2D datasets.

The anisotropy window has two modes: **Lumley triangle** and **barycentric map**. Both characterise the state of turbulence anisotropy at each point in the flow.

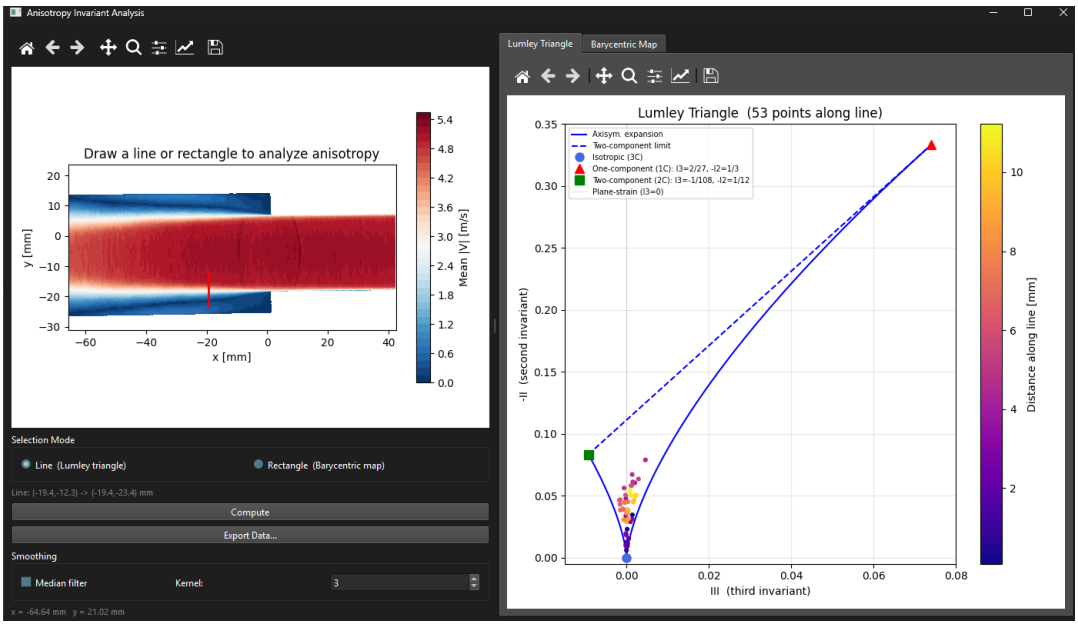

Figure 6.1: Anisotropy Invariant Analysis window. The left panel shows the mean field with the drawn reference line. The right panel shows the Lumley triangle with points coloured by distance along the line.

6.1 Lumley triangle mode

1. Ensure **Lumley Triangle** is selected.
2. Left-click and drag on the mean field (left panel) to draw a line.
3. The right panel shows the Lumley triangle with each point along your line plotted in a colour that represents its position along the line.

Points near the top vertex correspond to one-component turbulence. Points near the bottom-left correspond to two-component (axisymmetric contraction) and bottom-right to two-component axisymmetric expansion. Points near the centre are closer to isotropic.

6.2 Barycentric map mode

1. Select **Barycentric Map**.
2. Right-click and drag on the field to draw a rectangular ROI.
3. The right panel shows an RGB colour map overlaid on the flow domain, where red, green, and blue represent the three limiting states of anisotropy.

A **Median filter** checkbox applies a 3×3 spatial median filter to reduce noise in the anisotropy map.

Chapter 7

Correlation Analysis

Open this window from **4. Analysis → Correlation Analysis**.

This window computes two-point spatial correlations and (for TR data) temporal autocorrelations, and extracts integral length and time scales.

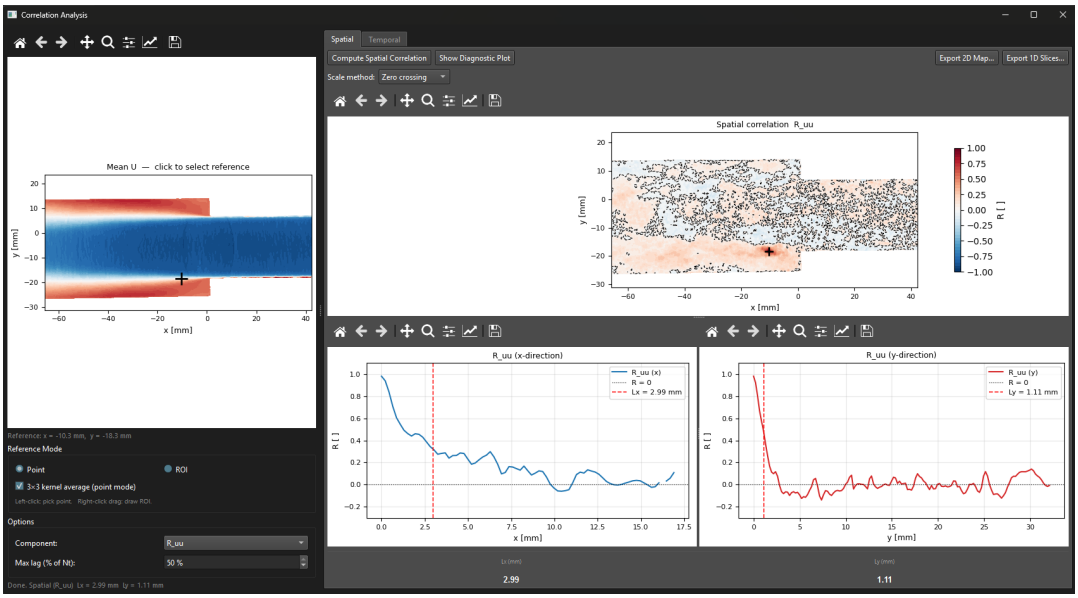

Figure 7.1: Correlation Analysis window in point mode. Top right: 2D spatial correlation map R_{uu}. Bottom: 1D slices in the x- and y-directions with the computed integral length scales L_x and L_y marked.

7.1 Reference mode

- **Point**: left-click on the mean field to select a reference point. Optionally enable **3 × 3 kernel average** to average the reference signal over a 3 × 3 patch of vectors, which reduces noise.
- **ROI**: right-click and drag to draw a rectangular region. The correlation is computed as the spatial average over all points within the ROI, following the method of Jose et al. (2026).

7.2 Spatial correlation (Spatial tab)

1. Select a reference point or ROI.
2. Choose the velocity component from the **Component** dropdown (R_{uu}, R_{vv}, R_{ww}).
3. Select the **Scale method** (see Section 7.2.1).

4. Click **Compute Spatial Correlation**.

In **point mode**, the result shows a 2D correlation map (top) and two 1D slices (bottom): one along the x-direction and one along the y-direction. Each 1D plot shows a vertical dashed red line marking the computed integral length scale L.

In **ROI mode**, only the two 1D curves are shown (no 2D map).

Click **Show Diagnostic Plot** to open a separate window with the cumulative integral of $R(\Delta r)$, which shows how the integral converges with separation distance.

7.2.1 Scale methods

Table 7.1: Integral scale methods available in uPrime.

Method	Description
Zero crossing	Integrates $R(\Delta r)$ from 0 to the first sustained zero crossing. Standard choice for clean correlations.
Exponential fit	Fits $R = e^{-\Delta r/L}$ and returns the fitted L. Robust when the correlation is noisy beyond the first lobe.
1/e point	Integrates $R(\Delta r)$ from 0 to the point where R first drops to $1/e \approx 0.368$.
Domain integral	Same as zero crossing but integrates to the full domain if no zero crossing is found.

The computed L_x and L_y values are shown in the readout strip at the bottom of the spatial tab.

7.3 Temporal correlation (Temporal tab, TR only)

1. Select a reference point or ROI using the same controls as the spatial tab.
2. Set **Max lag (% of Nt)** to control how far in time the correlation is computed. 50% is usually sufficient.
3. Select the **Scale method** for the integral time scale.
4. Click **Compute Temporal Correlation**.

The result shows $R(\tau)$ vs time lag τ in milliseconds, with the integral time scale T and Taylor microscale λ_t marked and displayed in the readout strip below the plot.

7.4 Exporting

- **Export 2D Map...**: saves the 2D correlation field (point mode only).
- **Export 1D Slices...**: saves the $R(\Delta x)$ and $R(\Delta y)$ curves as CSV.
- **Export R(τ)...**: saves the temporal autocorrelation curve as CSV.

Chapter 8

POD Analysis

Open this window from **4. Analysis → POD Analysis**.

Proper Orthogonal Decomposition (POD) decomposes the velocity field into a set of energy-ranked spatial modes and their associated temporal coefficients. The first few modes capture the most energetic structures in the flow.

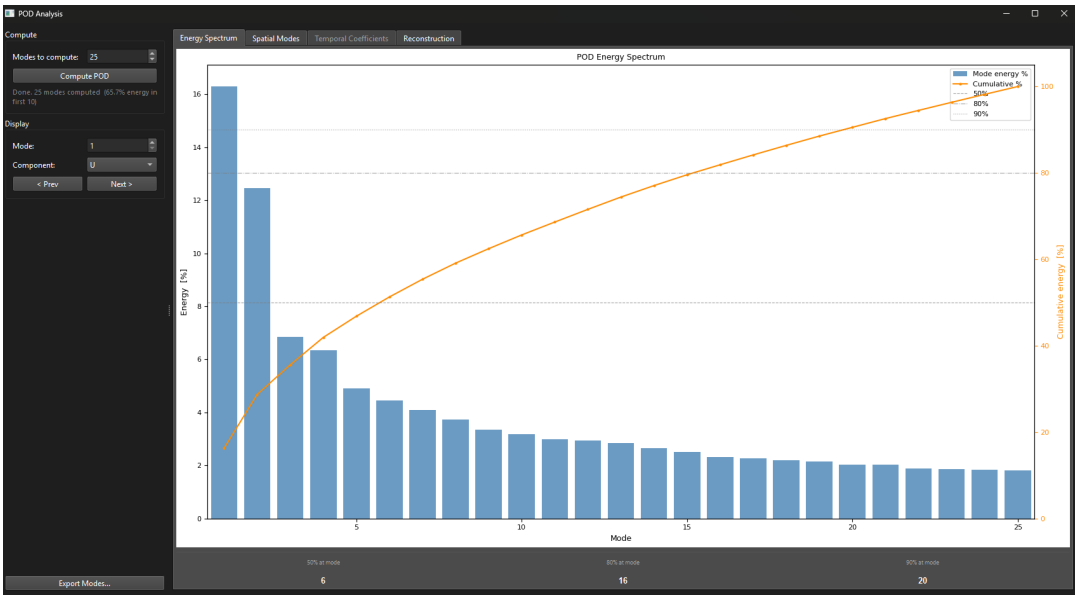

Figure 8.1: POD Energy Spectrum tab. Bar chart shows modal energy percentage; the orange curve shows cumulative energy. Dashed lines mark the 50%, 80%, and 90% thresholds. The readout strip at the bottom shows the mode count required for each threshold.

8.1 Computing POD

1. Set **Modes to compute** in the left panel. The default is 25. You can increase this but computation time grows with the number of modes.
2. Click **Compute POD**. The status label shows progress. For large datasets this may take several seconds.
3. Once complete, all four tabs are populated automatically.

> **Note**
>
> POD uses all available velocity components simultaneously (U and V for 2D data; U, V, and W for stereo). The energy ranking reflects the combined energy across all components.

8.2 Energy Spectrum tab

Shows a bar chart of the percentage of total TKE captured by each mode, with a cumulative energy curve overlaid on the right axis. Horizontal dashed lines mark the 50%, 80%, and 90% energy thresholds. The readout strip below the chart shows how many modes are needed to reach each threshold.

Tip

A steep energy drop after the first few modes indicates a flow dominated by coherent structures. A gradual decay indicates broadband turbulence.

8.3 Spatial Modes tab

Displays the spatial structure of a selected POD mode as a filled contour map (Figure 8.2).

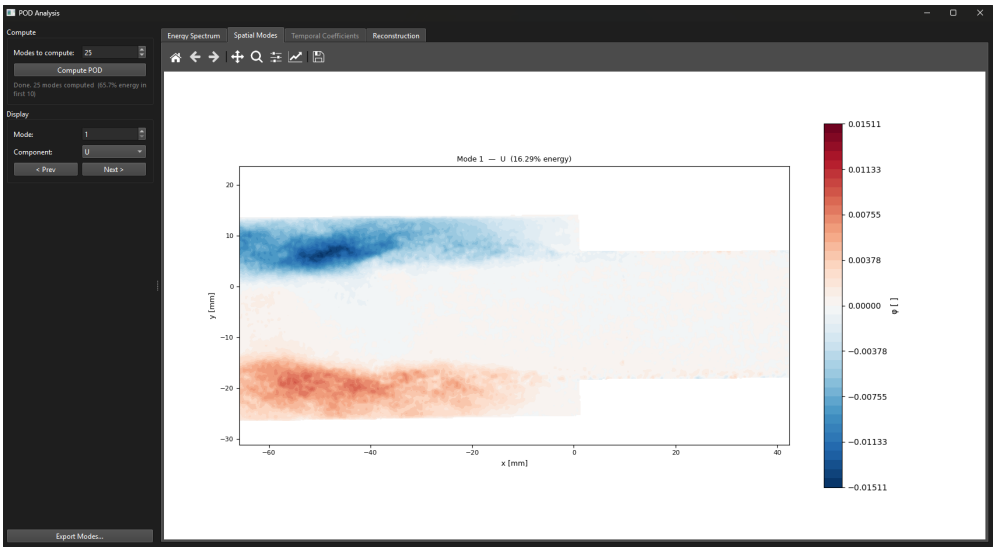

Figure 8.2: POD Spatial Modes tab showing Mode 1 of the U component (16.29% energy). Red and blue regions indicate opposite-signed velocity fluctuations associated with the dominant coherent structure.

- Use the **Mode** spinbox or the **< Prev** / **Next >** buttons to browse through modes.
- Use the **Component** dropdown (U / V / W) to switch the displayed velocity component of the mode.
- The title shows the mode number and the percentage of total energy it contains.

8.4 Temporal Coefficients tab (TR only)

Shows the time series $a_n(t)$ of the selected mode's temporal coefficient and its power spectral density (PSD). The PSD reveals the dominant frequency of the mode, which is useful for identifying periodic shedding or flapping.

8.5 Reconstruction tab

Shows a side-by-side comparison of an original snapshot and a low-order reconstruction using the first N modes (Figure 8.3).

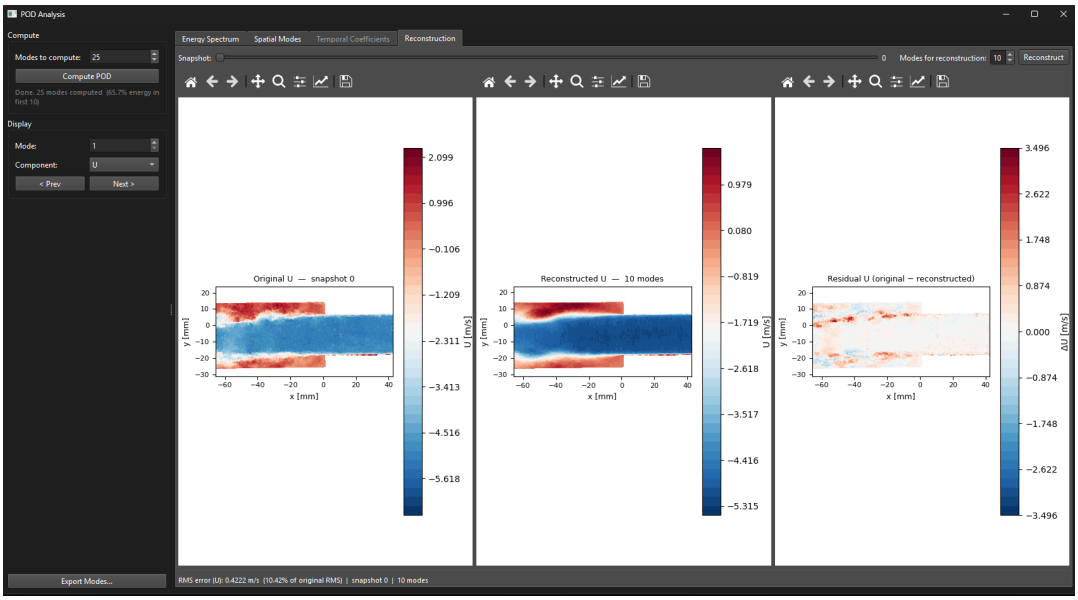

Figure 8.3: POD Reconstruction tab showing (left) an original snapshot, (centre) the reconstruction using 10 modes, and (right) the residual field. The RMS error is reported at the bottom of the window.

1. Drag the **Snapshot** slider to select which snapshot to display.
2. Set **Modes for reconstruction** to the number of modes to include.
3. Click **Reconstruct**.

Three panels are shown: the original snapshot, the reconstructed field, and the residual (original minus reconstructed). The RMS of the residual as a percentage of the original RMS is displayed below.

Tip

A small residual (below ∼10–20%) indicates that the selected number of modes captures most of the instantaneous structure. A large residual means more modes are needed, or the snapshot is dominated by small-scale turbulence that POD spreads across many modes.

8.6 Exporting

Click **Export Modes...** to save:

- All spatial modes as a Tecplot `.dat` file (one zone per mode).
- Temporal coefficients as a `.csv` file (one column per mode).

Chapter 9

Coordinate Transform

Open this tool from **5. Align / Transform → Transform / Align...**.

In experimental PIV, cameras are rarely perfectly aligned with the flow geometry. Even a small tilt in the camera mount or calibration target introduces a systematic angular error that propagates into every derived quantity — mean profiles, Reynolds stresses, line integrals, and integral scales — if left uncorrected. The uPrime transform tool is designed to fix this before any analysis is performed.

Two operations are available: **rotation** and **origin shift**. Both modify the dataset in memory and are reflected immediately in the main window field viewer.

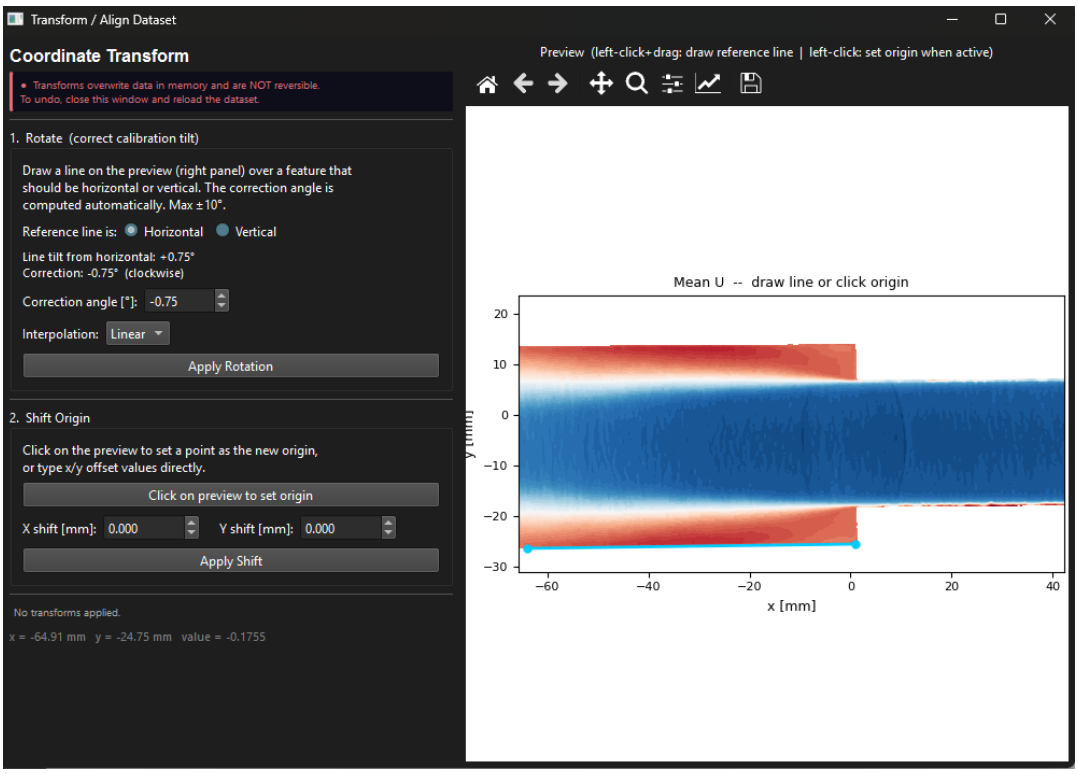

Figure 9.1: Coordinate Transform window. A reference line has been drawn along the lower pipe wall (cyan line, right panel). The correction angle is computed automatically and shown in the left panel before applying.

Warning

Transforms cannot be undone. If you need to try a different angle, reload the data and apply a fresh transform.

9.1 Rotation

1. Left-click and drag on the mean field preview to draw a reference line along any physical feature that should be perfectly horizontal or vertical — a pipe wall, step edge, free surface, or centreline.
2. uPrime computes the misalignment angle automatically and displays it in the **Angle** field. You can also type an angle directly if you know it.
3. Choose **Linear** or **Cubic** interpolation. Cubic gives smoother results and is recommended for final analysis; linear is faster for a quick check.
4. Click **Apply Rotation**. A progress bar tracks the operation. For large datasets the rotation runs in a background thread so the interface stays responsive.

Note

Rotation is limited to $\pm 10°$. This covers the typical range of camera mounting error. For larger corrections, revisit the calibration in DaVis before exporting.

Tip

Draw the reference line along the longest visible physical feature for the most accurate angle estimate. A step edge or channel wall spanning the full field width is ideal.

9.2 Origin shift

Moves the coordinate origin to a physically meaningful location — for example, a step corner, pipe inlet, or any reference point that makes downstream distances interpretable.

1. Click on the field preview at the desired new origin, **or** type the x and y values directly into the spinboxes.
2. Click **Apply Shift**.

9.3 Transform status

The main window shows a persistent status strip confirming every transform that has been applied, for example:

$$\textit{Data transformed: Rotated } -1.3° \;\cdot\; \textit{Shifted to } (51.0,\ 0.0)\ mm$$

This strip is visible at all times so you always know the state of the coordinate system before interpreting results.

Chapter 10

Tips and Best Practices

10.1 Data quality

- Ensure your DaVis export includes the valid vector mask. uPrime uses this mask to exclude spurious vectors from all calculations.
- For statistical convergence, use at least 2000 snapshots for non-TR data and at least 2 seconds of recording for TR data. uPrime warns you if these thresholds are not met.
- Apply the coordinate transform *before* opening any analysis window. All analysis windows use the coordinate system as it is at the time they are opened.

10.2 Workflow recommendation

A typical analysis session follows this order:

1. Load data.
2. Set acquisition type and f_s.
3. Apply rotation and origin shift if needed.
4. Inspect the mean field and verify the coordinate system looks correct.
5. Open Reynolds Stresses to check turbulence intensity and identify key regions.
6. Open TKE Budget to understand energy balance.
7. Open Correlation Analysis to extract integral length scales.
8. Open POD to identify dominant coherent structures.
9. Export results for figures and further analysis.

10.3 Performance

- POD computation scales with the number of snapshots squared. For datasets with more than 500 snapshots, limit the number of modes to 25–50 to keep computation times reasonable.
- Space–time spectral analysis and temporal correlations are the most memory-intensive operations. Close unused analysis windows to free RAM.
- On large grids ($> 500 \times 500$ vectors), the rotation transform may take 30–60 seconds. Use linear interpolation for a quick check and cubic for the final result.

10.4 Known limitations

- uPrime does not support 3D (tomographic) PIV data.
- Pressure fields and viscous dissipation cannot be computed from planar PIV alone; the TKE budget residual implicitly contains these terms.
- The Taylor hypothesis (required for converting temporal to spatial scales) is only valid when the turbulence intensity is low ($u'/\langle U \rangle \ll 1$). Use temporal scales directly when this condition is not met.
- Anisotropy analysis requires all three velocity components and is therefore stereo-only.

Chapter 11

Keyboard and Mouse Reference

11.1 Mouse interactions

Table 11.1: Mouse interactions common to all analysis windows.

Action	Effect
Left-click on field	Pick a reference point (correlation, spectra) or start drawing a line (Reynolds, TKE)
Left-click drag	Draw a free line profile
Right-click drag	Draw a rectangular ROI
Scroll wheel	Zoom in/out (when zoom tool is active in toolbar)
Middle-click drag	Pan (when pan tool is active in toolbar)

11.2 Toolbar buttons

Each plot panel has a toolbar with the following buttons (left to right):

Table 11.2: Matplotlib toolbar buttons.

Icon	Function
Home	Reset view to original zoom
Back / Forward	Navigate zoom history
Pan	Click and drag to pan
Zoom	Click and drag to zoom into a rectangle
Subplots	Adjust margins
Customise	Edit axis labels and limits
Save	Save the current figure as PNG, PDF, SVG, etc.

Note

While the zoom or pan tool is active, mouse clicks on the field do *not* trigger line drawing or point picking. Click the home or zoom button again to deactivate the toolbar tool before interacting with the data.

Appendix A

Citing uPrime

If you use uPrime in your research, please cite it as:

> Jibu Tom Jose. *uPrime: Open-source PIV post-processing toolkit.* Transient Fluid Mechanics Laboratory, Technion, 2026. https://doi.org/10.5281/zenodo.19376184

Appendix B

Version history

Version	Changes
v0.1	Initial release: Reynolds stresses, TKE viewer, basic spectra.
v0.2	Added TKE budget, anisotropy invariants, correlation analysis, coordinate transform. Migrated from PyQt5 to PyQt6.
v0.3	Added POD analysis with energy spectrum, spatial modes, temporal coefficients, and reconstruction. Improved correlation module with multiple scale methods and diagnostic plots.